Project Title

Multi-Pattern Unicode Search Engine Using Aho–Corasick Algorithm

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ABSTRACT

Multi-Pattern Unicode Search Engine

The **Multi-Pattern Unicode Search Engine** is a software system designed to efficiently search for **multiple keywords simultaneously** within large text datasets. The main purpose of the system is to identify and report every occurrence of a set of keywords in an input text while minimizing the time required for pattern matching.

The system uses the **Aho–Corasick string matching algorithm**, which combines a **Trie data structure** with **failure links** to efficiently process multiple patterns. Instead of searching for each keyword separately, the algorithm constructs a single pattern-matching structure and scans the input text in one traversal. This significantly improves the efficiency of searching, especially when thousands of keywords need to be matched against large text inputs.

The **Trie structure** stores all the keywords character by character, while **failure links** determine the next suitable state when a mismatch occurs during searching. The system also maintains output information at each relevant node to identify the keywords that have been matched. As a result, the system can report **every occurrence of each keyword**, including its position in the input text.

The system is designed to support **Unicode characters**, allowing it to process multilingual datasets containing different languages and character sets. This makes the search engine suitable for applications where text may contain English, Indian languages, Asian languages, special symbols, and other Unicode-supported characters.

The proposed system can process **large collections of keywords and text efficiently** and is designed to minimize unnecessary character comparisons. The performance of the system can be evaluated using parameters such as **search time, number of pattern matches, memory usage, and overall processing efficiency**. The algorithm provides an efficient alternative to repeatedly applying traditional single-pattern searching techniques.

The system has practical applications in **search engines, intrusion detection systems, spam and malware detection, plagiarism checking, keyword monitoring, text analysis, and multilingual information retrieval**. By combining the Trie data structure, failure links, and Unicode support, the proposed system provides a **fast, scalable, and efficient solution for multi-pattern keyword searching** across large and diverse text datasets.

Signature of the Guide